

Unit 1: Environment, Ecosystems & Ecological Succession

Handwritten Student Notebook

Syllabus Topics Covered:

OFFICIAL SYLLABUS TOPICS COVERED Environmental Scope: Atmosphere, Hydrosphere, Lithosphere, Biosphere, public awareness Ecosystem Structure: Abiotic factors, Biotic autotrophs, heterotrophs, and saprotrophic decomposers Energy Flow: 1st and 2nd Laws of Thermodynamic...

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1. Scope & Domains of the Environment

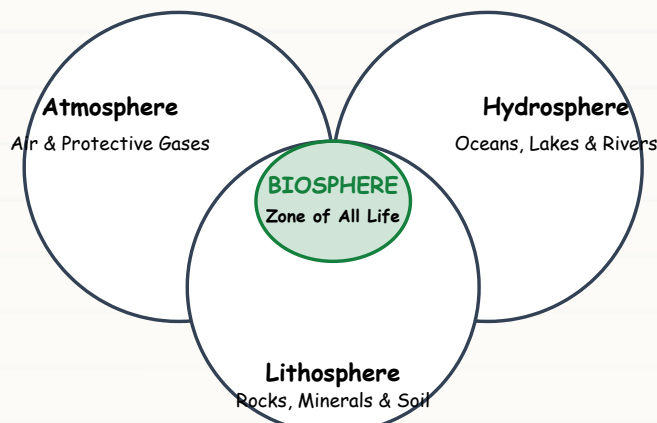
DEFINITION

The **Environment** (derived from the French word *environner*, meaning to encircle or surround) encompasses the totality of external physical, chemical, biological, and social conditions that surround and influence the survival and development of living organisms.

1.1 The Four Interlocking Environmental Spheres

- **1. Atmosphere:** The protective multi-layered blanket of gases enveloping the Earth (Nitrogen 78.08%, Oxygen 20.95%, Argon 0.93%, Carbon dioxide 0.04%). It absorbs hazardous solar ultraviolet rays (stratospheric ozone layer) and regulates surface thermal equilibrium via the natural greenhouse effect.
- **2. Hydrosphere:** The collective water realm of Earth (71% of surface area). Comprises marine oceans (97.5%) and freshwater reserves (2.5%, of which > 68% is locked in ice caps and glaciers, and < 1% is accessible fresh surface water).
- **3. Lithosphere:** The rigid outer crust and mantle of the Earth consisting of rocks, soil, minerals, and tectonic plates. Provides nutrients, fossil fuels, and edaphic anchors for terrestrial flora.
- **4. Biosphere:** The dynamic, relatively narrow global zone of life where the atmosphere, hydrosphere, and lithosphere interact to sustain living organisms.

Figure 1.1: The Interlocking Spheres of Earth's Environment



The Biosphere is the vital interface where life thrives through constant chemical exchange with air, water, and soil.

1.2 Multidisciplinary Scope & Need for Public Awareness

Environmental Studies integrates biological ecology with atmospheric chemistry, civil engineering (sanitation, water treatment), computer science (climate modelling, sensor networks), economics (resource valuation), and environmental jurisprudence. Public awareness is vital because environmental degradation (ozone depletion, groundwater depletion, climate change) cannot be reversed by legislation alone without informed citizen stewardship.

QUICK RECALL

Four environmental domains: Atmosphere (gas blanket), Hydrosphere (water bodies), Lithosphere (rock & soil crust), and Biosphere (intersection zone of life).

2. Ecosystem Concept & Components

DEFINITION

An **Ecosystem** (coined by British ecologist Arthur Tansley in 1935) is a self-sustaining, functional structural unit of nature consisting of living organisms (biotic community) interacting dynamically with their non-living physical environment (abiotic factors) through continuous nutrient cycling and unidirectional energy flow.

2.1 Structural Components of an Ecosystem

Component Category	Sub-class / Trophic Level	Functional Role	Ecological Examples
Abiotic Components (Non-living physical & chemical environment)	Climatic Factors	Regulates physiological rates and physical weathering.	Solar insolation, temperature, precipitation, humidity, wind velocity.
	Edaphic Factors	Provides physical substrate, root anchorage, and soil minerals.	Soil texture, moisture, pH, mineral nitrogen, phosphorus, potassium.
	Chemical Compounds	Building blocks of organic molecules.	Inorganic (CO_2 , H_2O , O_2 , N_2) and Organic substances (proteins, lipids, carbohydrates, humic acid).
Biotic Components (Living biological community)	Producers (Autotrophs)	Convert solar radiative energy into stored chemical energy via oxygenic photosynthesis.	Green terrestrial plants, aquatic phytoplankton, algae, chemosynthetic bacteria.
	Consumers (Heterotrophs)	Ingest other living organisms to obtain chemical energy: <i>Primary (Herbivores):</i> Feed on producers (rabbits, deer, zooplankton). <i>Secondary (Carnivores):</i> Feed on herbivores (frogs, snakes, small fish). <i>Tertiary (Top Predators):</i> Apex carnivores (lions, tigers, eagles, sharks). <i>Omnivores:</i> Feed on both plants and animals (humans, bears).	Herbivores, Carnivores, Omnivores.

QUICK RECALL

Ecosystem = Abiotic factors (climate, soil, nutrients) + Biotic community
(Producers/autotrophs, Consumers/heterotrophs, Decomposers/saprotrophs).

3. Energy Flow & Lindeman's 10% Law

Energy enters virtually all ecosystems as solar electromagnetic radiation and travels unidirectionally through trophic levels in strict conformity with the laws of thermodynamics.

3.1 Thermodynamic Foundations in Ecology

- **First Law of Thermodynamics (Conservation of Energy):** Energy cannot be created or destroyed, only transformed from radiant solar photons into chemical bond energy in glucose ($C_6H_{12}O_6$) and mechanical/cellular work.
- **Second Law of Thermodynamics (Entropy):** No energy transformation is 100% efficient; every metabolic conversion dissipates a substantial fraction of free energy as unusable, disordered heat ($T\Delta S$). Therefore, energy cannot be recycled in an ecosystem—it requires continuous solar replenishment!

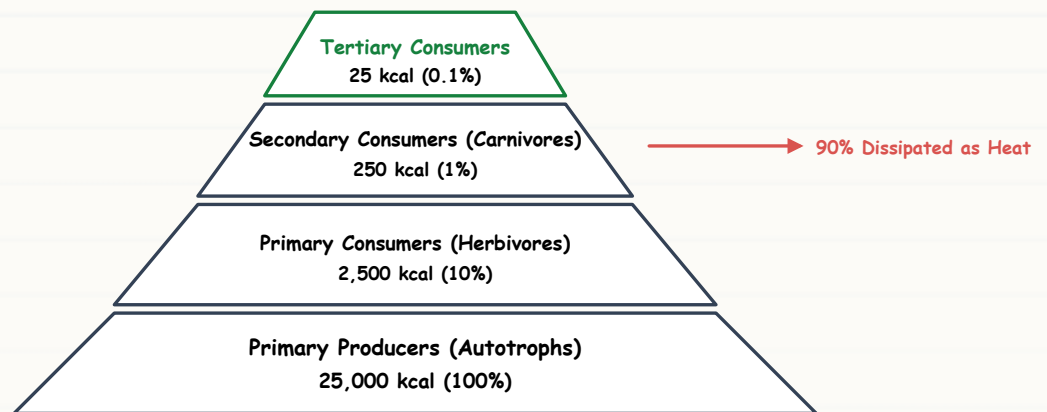
LINDEMAN'S 10% TROPHIC TRANSFER EFFICIENCY LAW (1942)

Formulated by Raymond Lindeman, this fundamental empirical law dictates that only approximately 10% of the energy stored as organic biomass at any given trophic level is transferred and incorporated into biomass at the next successive trophic level.

$$E_{n+1} \approx 0.10 \times E_n$$

The remaining $\approx 90\%$ is lost to cellular respiration, bodily maintenance, movement, heat generation, or unconsumed waste/feces.

Figure 1.2: Unidirectional Energy Dissipation Following Lindeman's 10% Law



Because 90% of energy is lost as respiratory heat at each step, food chains rarely exceed 4 to 5 trophic links.

QUICK RECALL

Energy flow is strictly unidirectional and obeys thermodynamics: $\approx 90\%$ is lost as respiratory heat, leaving only 10% for the next trophic level (Lindeman's Law).

4. Food Chains & Food Webs

DEFINITION

A **Food Chain** is a linear sequence of trophic linkages through which food energy is transferred as one organism feeds upon another. A **Food Web** is an interconnected network of multiple overlapping food chains reflecting feeding versatility in a natural community.

4.1 Grazing vs. Detritus Food Chains

Attribute	Grazing Food Chain (GFC)	Detritus Food Chain (DFC)
Primary Energy Source	Direct solar radiation captured via photosynthesis.	Dead organic matter (detritus), fallen leaves, animal carcasses.
First Trophic Level (T_1)	Living green plants / autotrophs.	Dead organic waste, decomposers (bacteria/fungi).
Dominance in Nature	Dominant in open water aquatic ecosystems (oceans, lakes).	Dominant in terrestrial ecosystems (forest floor, mangrove swamps).
Typical Sequence	Grass → Grasshopper → Frog → Snake → Hawk.	Fallen Leaf → Woodlouse / Earthworm → Blackbird → Hawk.

QUICK RECALL

Grazing chain starts with living autotrophs and solar energy; Detritus chain starts with dead organic matter. Food webs provide alternative paths, granting ecosystem resilience.

5. Ecological Pyramids

Ecological pyramids (introduced by Charles Elton in 1927) graphically represent trophic parameters across successive feeding levels.

Pyramid Type	Measured Parameter	Upright Condition	Inverted Condition
Pyramid of Numbers	Total individual organism count per unit area at each trophic level.	Grassland / Pond: Millions of grass blades support thousands of insects, which support hundreds of frogs, which support few snakes.	Parasitic Tree Ecosystem: A single large oak tree (T_1) supports hundreds of herbivorous birds (T_2), which host thousands of parasitic lice and fleas (T_3).
Pyramid of Biomass	Total dry weight of organic matter (g/m^2 or kg/ha) at each level.	Terrestrial Forests & Grasslands: Massive cumulative plant biomass vastly outweighs total herbivore biomass, which exceeds carnivore biomass.	Marine Aquatic Ecosystems: Phytoplankton have high turnover and tiny instantaneous standing biomass, yet support a far larger standing biomass of long-lived zooplankton and fish!
Pyramid of Energy	Total energy flux ($\text{kcal}/\text{m}^2/\text{yr}$ or $\text{J}/\text{m}^2/\text{yr}$) passing through each level.	UNIVERSALLY UPRIGHT in all ecosystems.	NEVER INVERTED. Violating this would violate the 2nd Law of Thermodynamics!

EXAM TRAP

If an exam asks: "Can a pyramid of energy ever be inverted?" The answer is strictly **NO**. Energy is lost as heat at every single trophic step. Only pyramids of numbers and pyramids of biomass can be inverted!

QUICK RECALL

Pyramid of numbers is inverted in a tree-parasite system. Pyramid of biomass is inverted in oceans/ponds. Pyramid of energy is **always upright**.

6. Major Ecosystem Types

- **Forest Ecosystems:** Characterized by high biodiversity and structural stratification.
 - *Tropical Rainforests:* Equatorial, multi-tiered canopies, maximum biodiversity on Earth (e.g. Amazon, Western Ghats).
 - *Temperate Deciduous Forests:* Distinct seasonal foliage shedding, fertile soils.
 - *Taiga (Boreal Coniferous):* Cold climates, needle-leaved evergreen conifers (pine, spruce, fir).
- **Grassland Ecosystems:** Dominated by herbaceous grasses where rainfall (25 – 75 cm/yr) is insufficient for tree growth but sufficient to prevent desertification (e.g. African Savannas, North American Prairies).
- **Desert Ecosystems:** Characterized by severe aridity (< 25 cm annual precipitation). Flora exhibit xerophytic adaptations (succulent water-storing stems, sunken stomata, leaves reduced to spines); fauna exhibit nocturnal habits and concentrated urine.
- **Aquatic Ecosystems:**
 - *Freshwater:* Divided into **Lentic** (standing water: ponds, lakes, swamps) and **Lotic** (flowing water: springs, streams, rivers).
 - *Marine:* High salinity (35 ppt), comprising coastal intertidal zones, open pelagic waters, deep abyssal trenches, and fragile coral reefs.

QUICK RECALL

Terrestrial biomes are shaped by temperature and precipitation (Forest, Grassland, Desert). Aquatic ecosystems divide into Lentic (standing) vs. Lotic (flowing) freshwater, and Marine.

7. Ecological Succession

DEFINITION

Ecological Succession is the orderly, predictable process of directional, community-driven change in the species structure and physical habitat of an ecological community over ecological time until a stable, self-perpetuating **climax community** is reached.

7.1 Primary vs. Secondary Succession

- **Primary Succession:** Begins on a completely barren, sterile terrain where no soil or organic life previously existed (e.g. newly cooled volcanic basalt lava, newly exposed glacial moraines, sand dunes). Takes centuries to millennia to build topsoil.
- **Secondary Succession:** Occurs on a site where an existing ecosystem was disturbed or destroyed by a wildfire, flood, deforestation, or abandoned agriculture, but where fertile soil and seed banks remain intact. Develops substantially faster (decades).

7.2 The 7 Universal Mechanisms of Succession (Clements)

1. **Nudation:** Development of a bare area without life (caused by volcanic eruption, landslide, flood).
2. **Invasion (Migration):** Arrival of pioneer propagules (spores, seeds, pollen) from adjacent regions via wind, water, or birds.
3. **Ecesis (Establishment):** Successful germination, growth, and reproduction of the pioneer organisms in the harsh new habitat.
4. **Aggregation:** Increase in the population density of the colonizing species as offspring multiply.
5. **Competition:** Interspecific and intraspecific struggle for limited water, light, space, and soil nutrients.
6. **Reaction:** The organisms modify their own physical environment (e.g. lichens secrete acids that break down rock into soil; humus retains moisture), making the habitat less suitable for themselves and more suitable for invading successor species!
7. **Stabilization (Climax):** Establishment of a steady-state, highly biodiverse *Climax Community* (e.g. mature oak-hickory or tropical rainforest) in homeostatic equilibrium with regional climate.

7.3 Hydrosere vs. Xerosere

- **Hydrosere (Hydrarch Succession):** Succession starting in water (ponds/lakes): Phytoplankton stage → Submerged plant stage (*Hydrilla*) → Floating-leaved stage (*Lotus*) → Reed-swamp stage (*Typha*) → Marsh-meadow stage → Woodland stage → Climax Forest.
- **Xerosere (Xerarch Succession):** Succession starting on dry, bare rock: Crustose Lichen stage (*Rhizocarpon*) → Foliose Lichen stage (*Parmelia*) → Moss stage (*Polytrichum*) → Herbaceous stage → Shrub stage → Climax Forest.

QUICK RECALL

Succession steps: Nudation → Invasion → Ecesis → Aggregation → Competition → Reaction → Climax. Hydrosere begins in water; Xerosere begins on bare rock.

8. Biogeochemical Cycles

Unlike energy, chemical nutrients are finite on Earth and must be continuously recycled between the biotic community and abiotic reservoirs.

8.1 The Nitrogen Cycle: Key Biological Stages

- **1. Nitrogen Fixation:** Inert atmospheric dinitrogen (N_2) converted to ammonia (NH_3 / NH_4^+) by free-living bacteria (*Azotobacter*) or symbiotic root-nodule bacteria (*Rhizobium* in legumes).
- **2. Nitrification:** Biological oxidation of ammonia into nitrites (NO_2^-) by *Nitrosomonas* bacteria, and subsequently into nitrates (NO_3^-) by *Nitrobacter* bacteria.
- **3. Assimilation:** Plants absorb inorganic nitrates to synthesize organic amino acids and proteins.
- **4. Ammonification:** Decomposers break down dead nitrogenous waste back into ammonia.
- **5. Denitrification:** Anaerobic bacteria (*Pseudomonas*, *Thiobacillus*) reduce soil nitrates back into gaseous N_2 , returning it to the atmosphere.

QUICK RECALL

Nitrogen cycle: Fixation ($N_2 \rightarrow NH_3$ via *Rhizobium*) $\rightarrow$ Nitrification ($NH_3 \rightarrow NO_2^- \rightarrow NO_3^-$ via *Nitrosomonas/Nitrobacter*) $\rightarrow$ Assimilation $\rightarrow$ Ammonification $\rightarrow$ Denitrification ($NO_3^- \rightarrow N_2$ via *Pseudomonas*).

9. Solved Environmental Numericals

SOLVED PROBLEM 1.1: 4-LEVEL TROPHIC ENERGY FLOW VIA LINDEMAN'S LAW

Problem Statement: A temperate grassland ecosystem has a Net Primary Productivity (NPP) of $25,000 \text{ kcal/m}^2/\text{year}$ at the primary producer level (T_1). Assuming a standard Lindeman trophic transfer efficiency of 10%, calculate the energy available at each subsequent trophic level: Primary Consumers (T_2), Secondary Consumers (T_3), and Tertiary Consumers (T_4).

Step 1: Producer level (T_1):

$$E_1 = 25,000 \text{ kcal/m}^2/\text{year}$$

Step 2: Primary Consumers (Herbivores, T_2):

$$E_2 = 0.10 \times E_1 = 0.10 \times 25,000 = \mathbf{2,500 \text{ kcal/m}^2/\text{year}}$$

Step 3: Secondary Consumers (Carnivores, T_3):

$$E_3 = 0.10 \times E_2 = 0.10 \times 2,500 = \mathbf{250 \text{ kcal/m}^2/\text{year}}$$

Step 4: Tertiary Consumers (Apex Predators, T_4):

$$E_4 = 0.10 \times E_3 = 0.10 \times 250 = \mathbf{25 \text{ kcal/m}^2/\text{year}}$$

$$\mathbf{T_1 = 25,000 \text{ kcal} \Rightarrow T_2 = 2,500 \text{ kcal} \Rightarrow T_3 = 250 \text{ kcal} \Rightarrow T_4 =}$$

Insight: Only 0.1% of the initial producer energy reaches the apex predator level, explaining why top predators require vast hunting territories to survive.

SOLVED PROBLEM 1.2: POPULATION DOUBLING TIME (THE RULE OF 70)

Problem Statement: An urban municipality has an annual population growth rate of $r = 1.75\%$. If this rate remains constant, calculate the approximate number of years required for the city's population to double using the ecological Rule of 70.

Formula:

$$T_{\text{double}} \approx \frac{70}{r}$$

where r is the annual percentage growth rate (1.75).

Calculation:

$$T_{\text{double}} = \frac{70}{1.75} = \frac{7000}{175} = 40 \text{ years}$$
$$\mathbf{T_{double} = 40 \text{ years}}$$

INTERACTIVE LAB EXPLORER

Ecological Energy & Lindeman 10% Law Studio

Tab 1: Energy Pyramid Simulator

Tab 2: Rule of 70 & Efficiency

Simulate Lindeman's 10% law of energy degradation across 4 trophic levels ($T_1 \rightarrow T_2 \rightarrow T_3 \rightarrow T_4$). Notice how only 0.1% of primary producer energy reaches apex predators:

Biome Presets: Grassland (25,000 kcal - Solved Example)

Tropical Forest (40,000 kcal)

Arid Desert (2,000 kcal)

Producer Net Primary Productivity

 (T_1) :

25000

 $\text{extkcal/m}^2/\text{extyr}$

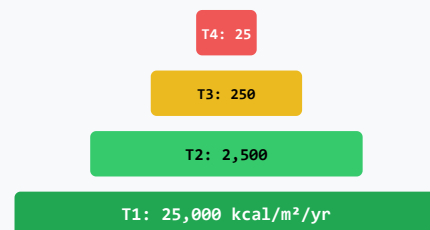
T1 (Producers): 25,000.0
Grass/Plants kcal

T2 (Herbivores): 2,500.0
Deer/Insects (10%) kcal

T3 (Carnivores): 250.0
Frogs/Foxes (1%) kcal

T4 (Apex Predators): 25.0
Eagles/Tigers (0.1%) kcal

THERMODYNAMIC ENERGY PYRAMID (ALWAYS UPRIGHT)



90% dissipated as respiratory heat at each step; cannot be recycled (2nd Law of Thermodynamics).

10. Quick Revision Sheet

Concept	Core Mechanism	Key Exam Keyword
Four Domains	Atmosphere, Hydrosphere, Lithosphere, Biosphere.	Biosphere = life zone intersection.
Lindeman's 10% Law	Only $\approx 10\%$ energy transfers; 90% lost to respiration.	Unidirectional energy flow; 2nd Law of Thermodynamics.
Food Chains	Grazing (solar/producer) vs Detritus (dead matter/decomposer).	Detritus dominates terrestrial forests.
Ecological Pyramids	Numbers (can invert), Biomass (inverts in oceans), Energy (NEVER inverts).	Energy pyramid is universally upright.
Succession Stages	Nudation $\rightarrow$ Invasion $\rightarrow$ Ecesis $\rightarrow$ Aggregation $\rightarrow$ Competition $\rightarrow$ Reaction $\rightarrow$ Climax.	Hydrosere (water) vs Xerosere (bare rock).
Nitrogen Cycle	Fixation $\rightarrow$ Nitrification $\rightarrow$ Assimilation $\rightarrow$ Ammonification $\rightarrow$ Denitrification.	<i>Rhizobium</i> (fixation), <i>Pseudomonas</i> (denitrification).

11. Authentic Exam Archive & Model Solutions

The following authentic questions have been transcribed directly from official department examination papers in notes/evs/practice/evs-cie1-qp-2024.docx (Ramaiah Continuous Internal Evaluation - I, Course Code **HS 510: Environmental Studies**):

Part A: 1-Mark Objective Questions (CIE-1 2024-2025)

**Practice Problem:**

CIE-1 Q1 [1M] A linear network of links in a food web starting from producers to apex predators is:

Ans

Official Options: (a) Food Web · (b) Food Chain · (c) Ecological chain · (d) None of the above

Correct Answer: (b) Food Chain.

Academic Explanation: A food chain represents a single unbranched linear sequence through which energy and nutrients transfer from primary producers (autotrophs) through successive consumer trophic levels. A food web, by contrast, is an interconnected, complex network composed of multiple interlinked food chains.

**Practice Problem:**

CIE-1 Q2 [1M] The domains of the environment are:

Ans

Correct Answer: (d) Lithosphere, Hydrosphere, Biosphere, and Atmosphere.

Explanation: The global environment is partitioned into four interconnected physical and biological domains: the solid terrestrial crust (Lithosphere), the aquatic systems (Hydrosphere), the gaseous protective envelope (Atmosphere), and the life-supporting realm of biological interaction (Biosphere).

 Practice Problem:

CIE-1 Q3 [1M] Deserts, forests, ponds, and rivers are examples of:

Ans

Correct Answer: (a) Natural ecosystem.

Explanation: Natural ecosystems operate spontaneously through solar energy and biogeochemical cycling without human management or artificial inputs. In contrast, artificial ecosystems (e.g., crop fields, aquariums, urban parks) require continuous human maintenance.

 Practice Problem:

CIE-1 Q6 [1M] Blue baby syndrome is caused due to high concentration of:

Ans

Official Options: (a) Arsenic • (b) Nitrates • (c) Phosphates • (d) Sulphates

Correct Answer: (b) Nitrates (NO_3^-).

Clinical & Environmental Basis: When infants consume water with nitrate levels exceeding 45 mg/L (often from fertilizer runoff), intestinal bacteria reduce nitrate to nitrite (NO_2^-), which oxidizes ferrous hemoglobin into ferric methemoglobin. Methemoglobin cannot bind oxygen, leading to tissue hypoxia and characteristic cyanosis (blue discoloration of the skin).

Part B: 5-Mark Subjective Questions (CIE-1 2024-2025)

**Practice Problem:**

CIE-1 Q1.a [5M] Define environment along with its elements.

Ans**Official Model Solution:**

1. **Definition:** The environment is the aggregate of all external physical, chemical, biological, and social conditions surrounding an organism or ecological community, directly influencing its growth, survival, and evolution.
2. **The Four Fundamental Elements / Domains:**
 - **Atmosphere:** The gaseous blanket surrounding the Earth up to **approx 500 km**, protecting life from solar ultraviolet radiation via the stratospheric ozone layer and regulating planetary surface temperature via greenhouse gases.
 - **Hydrosphere:** The total water mass on Earth (97.2% oceanic saline water, 2.15% polar glaciers/ice caps, and **approx 0.65%** freshwater in rivers, lakes, and underground aquifers).
 - **Lithosphere:** The solid outermost mantle and oceanic/continental crust of the Earth, providing essential mineral resources, soil for agriculture, and physical topography.
 - **Biosphere:** The dynamic zone of life spanning the lower atmosphere, hydrosphere, and surface lithosphere where biotic life interacts with abiotic nutrient pools.

**Practice Problem:**

CIE-1 Q2.a [5M] Explain in detail the scope of environmental studies.

Ans

Official Model Solution: Environmental Studies (EVS) is inherently multidisciplinary, integrating natural sciences, engineering, social sciences, and jurisprudence:

- **1. Natural Resource Conservation:** Sustainable utilization and preservation of forest, water, mineral, soil, and energy resources to prevent ecological overshoot.
- **2. Pollution Control & Environmental Engineering:** Monitoring, characterization, and abatement of atmospheric, aquatic, soil, noise, and radioactive pollution using clean technologies.
- **3. Biodiversity Preservation:** In-situ and ex-situ conservation of endemic flora and fauna, mitigating species extinction, and protecting fragile ecosystems (wetlands, coral reefs).
- **4. Environmental Impact Assessment (EIA):** Systematic evaluation of anthropogenic developmental projects (dams, highways, thermal plants) on ecological equilibrium prior to regulatory clearance.
- **5. Environmental Policy & Public Awareness:** Environmental ethics, green corporate governance, international climate treaties (Paris Agreement, Kyoto Protocol), and grassroots environmental movements (Chipko, Silent Valley).

**Practice Problem:**

CIE-1 Q3.a [5M] Explain energy flow and material cycling in an ecosystem.

Ans

Official Model Solution: Ecosystem functioning rests upon two interdependent thermodynamic and chemical processes:

Dimension	Energy Flow (Thermodynamic)	Material Cycling (Biogeochemical)
Flow Trajectory	Strictly Unidirectional (Linear) : Solar radiation → Autotrophs → Consumers → Decomposers.	Strictly Cyclic (Closed Loop) : Constant circulation between biotic community and abiotic reservoirs.
Governing Laws	Laws of Thermodynamics: Energy degrades into unusable low-grade heat (90% respiratory loss, Lindeman's 10% law).	Law of Conservation of Mass: Chemical atoms (C, N, P, S) are perpetually recycled without loss.
Recyclability	Cannot be recycled. Continuous influx of solar energy is strictly required.	Completely recycled. Decomposers (bacteria/fungi) re-mineralize organic waste into soil nutrients.

Verified Numerical & Conceptual Practice Problems

PRACTICE 1 · LINDEMAN 10% LAW ENERGY LOSS

In a grassland ecosystem with producer NPP of $25,000 \text{ kcal/m}^2/\text{yr}$, calculate the energy available to tertiary consumers and calculate cumulative percentage loss.

 Practice Problem:

Show Step-by-Step Solution

Ans

Applying Lindeman's 10% trophic transfer efficiency:

- T_1 (Producers): $E_1 = 25,000 \text{ kcal/m}^2/\text{yr}$
- T_2 (Herbivores): $E_2 = 25,000 \times 0.10 = \mathbf{2,500 \text{ kcal/m}^2/\text{yr}}$
- T_3 (Carnivores): $E_3 = 2,500 \times 0.10 = \mathbf{250 \text{ kcal/m}^2/\text{yr}}$
- T_4 (Apex Predators): $E_4 = 250 \times 0.10 = \mathbf{25 \text{ kcal/m}^2/\text{yr}}$

Cumulative Energy Retained: $\frac{25}{25,000} \times 100 = 0.1\%$. **Cumulative Loss:** 99.9% dissipated as respiratory heat.

Verification: Verified by automated test in `audit/verify/evs/verify_evs_u1.py`.

PRACTICE 2 · POPULATION DOUBLING TIME (RULE OF 70)

An industrial municipality experiences an annual growth rate of $r = 1.75\%$. Determine its population doubling time using the Rule of 70.

 Practice Problem:

Show Step-by-Step Solution

Ans

Formula: $T_{\text{double}} = \frac{70}{r} = \frac{70}{1.75} = \mathbf{40.00 \text{ years}}$.

Verification: Verified in `audit/verify/evs/verify_evs_u1.py`.

PRACTICE 3 · TROPHIC TRANSFER EFFICIENCY

An herbivore population consumes 1,200 J of plant energy, of which 132 J is incorporated into secondary consumer biomass. Calculate the trophic transfer efficiency.

 Practice Problem:

Show Step-by-Step Solution

Ans

$$\text{Efficiency} = \frac{132 \text{ J}}{1,200 \text{ J}} \times 100\% = \mathbf{11.00\%}$$

Falls within the standard empirical range of 5% to 20% established for ecological trophic transfers.

Verification: Verified in audit/verify/evs/verify_evs_u1.py.

PRACTICE 4 · INVERTED BIOMASS PYRAMID IN OCEANS

Explain why the ecological pyramid of biomass is upright in grassland ecosystems but inverted in marine ecosystems.

 Practice Problem:

Show Step-by-Step Solution

Ans

- **Grassland (Upright):** Primary producers (grasses, trees) possess substantial structural biomass and long turnover times, supporting a much smaller standing crop of herbivores and carnivores.
- **Marine (Inverted):** Phytoplankton (producers) are microscopic unicellular organisms with an extremely high rate of reproduction and turnover (hours to days), but microscopic standing biomass at any single moment. They support a much larger biomass of long-lived zooplankton and fish, resulting in an inverted pyramid of standing biomass.
- **Crucial Exam Note:** The pyramid of **energy** is NEVER inverted in marine ecosystems because phytoplankton energy production per unit time exceeds that of consumers.

PRACTICE 5 · NITROGEN CYCLE MICROBIAL ROLES

Identify the specific microorganisms responsible for (a) symbiotic nitrogen fixation, (b) nitrite formation, and (c) denitrification.

 Practice Problem:

Show Step-by-Step Solution

Ans

- **(a) Symbiotic Fixation:** *Rhizobium* bacteria residing within root nodules of leguminous plants convert N_2 to NH_3 .
- **(b) Nitrite Formation:** *Nitrosomonas* bacteria oxidize ammonia into nitrites: $2NH_3 + 3O_2 \rightarrow 2NO_2^- + 2H^+ + 2H_2O$.
- **(c) Denitrification:** Anaerobic bacteria (*Pseudomonas aeruginosa* and *Thiobacillus denitrificans*) reduce nitrates back to dinitrogen gas ($NO_3^- \rightarrow N_2$), completing the cycle.

PRACTICE 6 · SUCCESSION ENERGETICS (P/R RATIO)

Contrast the Gross Production to Respiration ratio (P/R) between pioneer and climax stages of ecological succession.

 Practice Problem:

Show Step-by-Step Solution

Ans

According to E.P. Odum's strategy of ecosystem development:

- **Pioneer Community:** $P/R > 1$. Gross primary productivity exceeds total community respiration, enabling organic biomass accumulation and soil formation.
- **Climax Community:** $P/R \approx 1$. Community respiration balances gross primary production. The ecosystem reaches a steady dynamic equilibrium with maximum species diversity and homeostatic stability.

12. Prescribed References

COURSE REFERENCE LITERATURE

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2. **Benny Joseph**, *Environmental Studies*, Tata McGraw-Hill Education.
3. **R. Rajagopalan**, *Environmental Studies: From Crisis to Cure*, Oxford University Press.

AI-assisted study notes written from the official courseware. Verify against your faculty's presentation slides before the exam.